Chronic Disease Data Transformation Assignment Using Spark

Course/Project Context: As part of building a US Chronic Disease Trends Dashboard, you will perform data transformations in PySpark to prepare CDC Chronic Disease Indicators data for visualization in Power BI. Focus on the medallion architecture (Silver to Gold layer transformations). Use the provided dataset (downloaded CSV from CDC) and assume a Spark session is set up.

Objectives:

- Apply Spark transformations to clean, aggregate, and model data.
- Ensure data quality and optimization.
- Prepare Gold-layer tables (e.g., Parquet) suitable for Power BI import (e.g., fact and dimension tables).

Prerequisites:

- Download the CDC dataset CSV.
- Set up PySpark environment (local or Databricks).
- Load raw data into a Bronze DataFrame: df_bronze = spark.read.csv("path_to_csv", header=True, inferSchema=True).

Assignment Structure: This assignment is divided into tasks aligned with the Power BI dashboard pages. For each task, write PySpark code to transform the data from Silver (cleaned) to Gold (modeled/aggregated). Include quality checks, optimizations, and output as partitioned Parquet. Submit your code snippets, explanations, and any assumptions.

Task 1: Transformations for Overview Page

Dashboard Requirement: US map with prevalence bubbles (e.g., diabetes % by county), top chronic conditions KPIs. **Data Needs:** Aggregated prevalence rates by location (county/state), condition, with KPIs like average prevalence.

1. Starting from a Silver DataFrame (assume df_silver with columns like 'LocationDesc', 'LocationID' (FIPS), 'Topic' (e.g., 'Diabetes'), 'DataValue' (prevalence %), 'YearStart', 'StratificationCategory' (e.g., 'Overall')): Write code to filter for key conditions (diabetes, heart disease, obesity). Aggregate average prevalence by county (groupBy 'LocationID', 'Topic'). Include a quality check to flag counties with missing DataValue > 20%.
2. Create a dimension table DimLocation from the data (select distinct 'LocationID', 'LocationDesc', 'StateAbbr'). How would you handle duplicates or nulls in location data?

3. Optimize the aggregation for large datasets: Use repartitioning or caching. What partition key would you use when writing to Parquet (e.g., by 'StateAbbr')?

Task 2: Transformations for Trends Over Time Page

Dashboard Requirement: Line charts for national/state changes in obesity, heart disease, etc. **Data Needs:** Time-series aggregations (e.g., yearly averages) at national and state levels.

1. From df_silver, filter for conditions like 'Obesity' and 'Cardiovascular Disease'. Group by 'YearStart', 'Topic', and 'LocationAbbr' (state) to compute yearly average DataValue. Add a calculated column for year-over-year change (using window functions).

2. Implement a quality check: Identify and handle years with incomplete data (e.g., count records per year < expected threshold). Use Spark SQL for this.

3. For national trends, aggregate further by 'YearStart' and 'Topic' (sum or avg across all locations). How would you ensure the data is sorted chronologically before writing to Gold Parquet?

Task 3: Transformations for Disparities Page

Dashboard Requirement: Breakdown by race/ethnicity, age, gender (stacked bars). **Data Needs:** Stratified aggregations showing prevalence by demographic groups.

1. Using df_silver, filter where 'StratificationCategory' is 'Race/Ethnicity', 'Age', or 'Gender'. Pivot the data to create columns for each stratification (e.g., prevalence by 'White', 'Black', etc.) grouped by 'Topic' and 'LocationAbbr'.

2. Add a quality validation: Check for consistency across stratifications (e.g., total overall should approximate sum of subgroups). Use assertions or custom UDFs.

3. Create a DimDemographic table (distinct 'StratificationCategory', 'Stratification'). What transformations would you apply to standardize values (e.g., title case)? Optimize by broadcasting small tables if joining.

Task 4: Transformations for Risk Factors Page

Dashboard Requirement: Correlations (e.g., physical inactivity vs. diabetes). **Data Needs:** Joined datasets showing relationships between risk factors and conditions.

1. Identify risk factor topics (e.g., 'Nutrition, Physical Activity, and Weight Status' for inactivity). Join df_silver on 'LocationID' and 'YearStart' between risk factors and conditions (e.g., inactivity prevalence joined with diabetes prevalence).
2. Compute a correlation metric (e.g., simple Pearson using Spark MLlib or approx calculation). Include a filter for sufficient data points per location.
3. Quality check: Handle mismatched joins (e.g., left join and fill nulls with averages). How would you optimize the join (e.g., broadcast if one side is small)? Write the output as a fact table with paired metrics.

Task 5: Transformations for Pipeline Overview Page

Dashboard Requirement: Diagram page explaining your Spark architecture and quality metrics. **Data Needs:** Metadata tables with pipeline stats (e.g., row counts, quality issues).

1. From your transformations above, create a metadata DataFrame logging stats (e.g., row count before/after each step, null percentages). Use Spark actions like count() and describe().
2. Aggregate quality metrics (e.g., number of flagged records per task). How would you union these from multiple DataFrames?
3. Output a simple Gold table for Power BI (e.g., columns: 'Task', 'Metric', 'Value'). What transformations ensure this is easily queryable?